Fully Connected Neural Network

Complete step-by-step forward pass, loss, backpropagation, gradients, and one parameter update

Architecture: 3 → 2 → 2 → 2 → 1 | Activation: Sigmoid | Loss: 1/2 squared error

1. Architecture

We choose 2 neurons in each of the three hidden layers.

```
a^(0) = x
z^(l) = W^(l)a^(l-1) + b^(l)
a^(l) = sigmoid(z^(l))
```

Layer	Connection	W dimension	b dimension
1	3 → 2	2 × 3	2 × 1
2	2 → 2	2 × 2	2 × 1
3	2 → 2	2 × 2	2 × 1
Output	2 → 1	1 × 2	1 × 1

2. Input, Target, and Initial Parameters

```
x = [1, 0.5, -1]^T
Target y = 0.7

W^(1) = [[0.10, -0.20, 0.05], [0.15, 0.10, -0.10]]
b^(1) = [0.01, -0.02]^T

W^(2) = [[0.20, -0.10], [0.05, 0.15]]
b^(2) = [0.02, -0.01]^T

W^(3) = [[0.10, 0.20], [-0.05, 0.10]]
b^(3) = [0.01, 0.02]^T

W^(4) = [[0.30, -0.20]]
b^(4) = [0.05]
```

3. Activation Function

```
sigmoid(z) = 1/(1 + e^(-z))
sigmoid'(z) = sigmoid(z)[1 - sigmoid(z)]
```

4. Forward Pass — Layer 1

Formula → substitute → result

```
z^(1) = W^(1)x + b^(1)
z1^(1) = (0.10)(1) + (-0.20)(0.5) + (0.05)(-1) + 0.01
= 0.10 - 0.10 - 0.05 + 0.01 = -0.04

z2^(1) = (0.15)(1) + (0.10)(0.5) + (-0.10)(-1) - 0.02
= 0.15 + 0.05 + 0.10 - 0.02 = 0.28
```

```
a1^(1) = sigmoid(-0.04) ≈ 0.4900
a2^(1) = sigmoid(0.28) ≈ 0.5695
a^(1) ≈ [0.4900, 0.5695]^T
```

5. Forward Pass — Layer 2

```
z^(2) = W^(2)a^(1) + b^(2)
z1^(2) = (0.20)(0.4900) + (-0.10)(0.5695) + 0.02 ≈ 0.06105
z2^(2) = (0.05)(0.4900) + (0.15)(0.5695) - 0.01 ≈ 0.09993
```

```
a1^(2) = sigmoid(0.06105) ≈ 0.5153
a2^(2) = sigmoid(0.09993) ≈ 0.5250
a^(2) ≈ [0.5153, 0.5250]^T
```

6. Forward Pass — Layer 3

```
z^(3) = W^(3)a^(2) + b^(3)
z1^(3) = (0.10)(0.5153) + (0.20)(0.5250) + 0.01 ≈ 0.16653
z2^(3) = (-0.05)(0.5153) + (0.10)(0.5250) + 0.02 ≈ 0.04673

a1^(3) = sigmoid(0.16653) ≈ 0.5415
a2^(3) = sigmoid(0.04673) ≈ 0.5117
a^(3) ≈ [0.5415, 0.5117]^T
```

7. Forward Pass — Output

```
z^(4) = W^(4)a^(3) + b^(4)
= (0.30)(0.5415) + (-0.20)(0.5117) + 0.05
≈ 0.11011

■ = sigmoid(0.11011) ≈ 0.5275
```

Prediction = 0.5275, while target = 0.7.

8. Loss

```
L = 1/2(■ - y)^2
= 1/2(0.5275 - 0.7)^2
≈ 0.01488
```

The factor 1/2 is used so that differentiating the square gives a cleaner expression.

9. Backpropagation Formulas

```
Define δ^(l) = ∂L/∂z^(l).

Output: δ^(4) = (■-y)sigmoid'(z^(4))
Hidden: δ^(l) = [W^(l+1)]^T δ^(l+1) ■ sigmoid'(z^(l))
Weight gradient: ∂L/∂W^(l) = δ^(l)[a^(l-1)]^T
Bias gradient: ∂L/∂b^(l) = δ^(l)
```

10. Output Layer — Error Signal

```
δ^(4) = (■-y)sigmoid'(z^(4))
■-y = 0.5275 - 0.7 = -0.1725
sigmoid'(z^(4)) = ■(1-■) = (0.5275)(0.4725) ≈ 0.2492

δ^(4) = (-0.1725)(0.2492) ≈ -0.0430
```

11. Output Weight and Bias Gradients

```
∂L/∂W^(4) = δ^(4)[a^(3)]^T
= (-0.0430)[0.5415, 0.5117]
≈ [-0.0233, -0.0220]

∂L/∂b^(4) = δ^(4) ≈ -0.0430
```

12. Hidden Layer 3 — Backpropagation

```
δ^(3) = [W^(4)]^Tδ^(4) ■ sigmoid'(z^(3))

[W^(4)]^Tδ^(4) = [[0.30],[-0.20]](-0.0430)
≈ [-0.0129, 0.0086]^T
```

```
sigmoid'(z1^(3)) = 0.5415(1-0.5415) ≈ 0.2483
sigmoid'(z2^(3)) = 0.5117(1-0.5117) ≈ 0.2499

δ^(3) ≈ [-0.00320, 0.00215]^T

∂L/∂W^(3) = δ^(3)[a^(2)]^T
≈ [[-0.00165, -0.00168], [0.00111, 0.00113]]
∂L/∂b^(3) ≈ [-0.00320, 0.00215]^T
```

13. Hidden Layer 2 — Backpropagation

```
δ^(2) = [W^(3)]^Tδ^(3) ■ sigmoid'(z^(2))
[W^(3)]^Tδ^(3) ≈ [-0.000428, -0.000425]^T

sigmoid'(z1^(2)) ≈ 0.2498
sigmoid'(z2^(2)) ≈ 0.2494

δ^(2) ≈ [-0.000107, -0.000106]^T

∂L/∂W^(2) = δ^(2)[a^(1)]^T
≈ [[-0.0000523, -0.0000608], [-0.0000520, -0.0000605]]
∂L/∂b^(2) ≈ [-0.000107, -0.000106]^T
```

14. Hidden Layer 1 — Backpropagation

```
δ^(1) = [W^(2)]^Tδ^(2) ■ sigmoid'(z^(1))
δ^(1) ≈ [-6.66×10^(-6), -1.28×10^(-6)]^T

∂L/∂W^(1) = δ^(1)x^T
≈ [[-6.66×10^(-6), -3.33×10^(-6), 6.66×10^(-6)],
[-1.28×10^(-6), -0.64×10^(-6), 1.28×10^(-6)]]

∂L/∂b^(1) ≈ [-6.66×10^(-6), -1.28×10^(-6)]^T
```

15. Complete Error Flow

```
δ^(4) ≈ [-0.0430]
↓
δ^(3) ≈ [-0.00320, 0.00215]^T
↓
δ^(2) ≈ [-0.000107, -0.000106]^T
↓
δ^(1) ≈ [-6.66×10^(-6), -1.28×10^(-6)]^T
```

This is the chain rule: the output error is propagated backward through transposed weight matrices and local activation derivatives.

16. One Gradient-Descent Update

```
W_new = W_old - η(∂L/∂W)
b_new = b_old - η(∂L/∂b)
η = 0.1
```

Output layer:

```
W^(4)_new = [0.30,-0.20] - 0.1[-0.0233,-0.0220]
≈ [0.30233,-0.19780]
b^(4)_new = 0.05 - 0.1(-0.0430) ≈ 0.05430
```

Layer 3:

```
W^(3)_new ≈ [[0.100165,0.200168],[-0.050111,0.099887]]
b^(3)_new ≈ [0.010320,0.019785]^T
```

Layer 2:

```
W^(2)_new ≈ [[0.2000052,-0.0999939],[0.0500052,0.1500060]]
b^(2)_new ≈ [0.0200107,-0.0099894]^T
```

Layer 1:

```
W^(1)_new ≈ [[0.1000007,-0.1999997,0.0499993],
[0.1500001,0.1000001,-0.1000001]]
b^(1)_new ≈ [0.0100007,-0.0199999]^T
```

17. Final Formula Sheet

```
FORWARD
z = Wa + b
a = sigmoid(z)

LOSS
L = 1/2(■-y)^2

BACKWARD
δ_output = (■-y)sigmoid'(z)
δ_hidden = W^Tδ_next ■ sigmoid'(z)
∂L/∂W = δa^T
∂L/∂b = δ

UPDATE
W ← W - η∂L/∂W
b ← b - η∂L/∂b
```